📖 SCAFFOLDED WORKBOOK · YEAR 9–10

ADVANCED WATER SAFETY

Health & Physical Education · Year 9–10 · Victorian Curriculum F–10 Version 2.0

This workbook has **word banks**, **sentence starters**, and **structured frameworks** to help you. The questions are the same as the standard workbook — the scaffolds just make it easier to get started.

STUDENT NAME

CLASS

TEACHER

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📌 How the scaffolds work

- **Word banks** — give you the key terms to choose from so you don't have to think of them yourself
- **Sentence starters** — begin the sentence for you; just finish it in your own words
- **Argument frameworks** — break an argument into parts (claim, evidence, explanation) so you can build it step by step
- **Calculation guides** — show you the formula and what to substitute in

1 Advanced Data Analysis — Victorian & Global Drowning Trends

📋 KEY DATA — KEEP THIS HANDY

📊 **52 people** drowned fatally in Victoria in 2024–25 — 9% above the ten-year average of 48.

🧮 Victoria's population ≈ **6.9 million**. Rate formula: deaths ÷ (population ÷ 100,000). So: 52 ÷ 69 = **0.75 per 100,000**.

🇦🇺 Australia's national drowning rate = **1.13 per 100,000** in 2025/26 (RLSSA). Victoria at 0.75 is BELOW the national rate.

🌏 People from multicultural backgrounds = **37%** of Victorian drowning deaths since 2015–16 (185 deaths). Country of birth was unknown in a further **22%** of cases, so 37% is a lower bound.

👨 **73%** of Victorian drowning deaths (38 of 52) were male.

🚫 **Zero deaths** in Victorian public swimming pools — the first time since 2019–20. But 25% of deaths were in home environments.

Activity 1.1 — Calculations & Interpretation ~15 min

Question 1.1a — Calculation 2 marks

Calculate the percentage of Victorian drowning deaths that were male. Show your working using the formula below.

CALCULATION FORMULA

Percentage = (number of males ÷ total deaths) × 100

= (_____ ÷ 54) × 100 = _____%

My answer: _____% of Victorian drowning deaths were male.

Now compare to the global figure of ~80%. Is Victoria higher or lower?

Victoria's rate is higher / lower (circle one) than the global figure of 80%.

This suggests that _____

Question 1.1b — Rate Comparison

3 marks

Calculate: if Victoria drowned at the national rate of 1.13 per 100,000 (2025/26), how many deaths would be expected? Compare with the 52 actually recorded. Show working.

CALCULATION GUIDE

Expected deaths = rate × (population ÷ 100,000)

= 1.2 × _____ (units of 100,000 in 6.9M) = _____ deaths

At the national rate, Victoria would expect approximately _____ deaths per year.

This is more / fewer (circle) than the actual figure of 52 deaths.

What this tells us about Victoria's performance: _____

Question 1.1c — CALD Analysis

3 marks

People from multicultural backgrounds account for 37% of Victorian drowning deaths since 2015–16, and 58% of fishing-related drowning deaths. Are they over-represented? Explain TWO structural reasons.

WORD BANK

over-represented | structural | swim lessons | language barriers | fishing risk | 40% > 28%

CALD Victorians are _____ in drowning deaths because 37% of deaths since 2015–16 were people from multicultural backgrounds.

Reason 1 — Use this argument framework:

REASON: _____

BECAUSE: _____

THIS MEANS: _____

Reason 2 — Use this argument framework:

REASON: _____

BECAUSE: _____

THIS MEANS: _____

2 Physiology of Drowning — What Happens in the Body

📋 KEY FACTS — PHYSIOLOGY

🧠 **4–6 minutes** — after this long without oxygenated blood, brain cells start dying irreversibly. This is called **cerebral hypoxia**.

💓 CPR generates about **25–30%** of normal cardiac output — enough to keep the brain alive until the heart can restart.

🥶 **1-10-1 Cold Water Rule:** ~1 minute to control breathing · ~10 minutes of useful swimming · up to 1 hour before hypothermia.

💧 **Secondary drowning** — fluid in the lungs can develop HOURS after a near-drowning. Always get a medical check after any submersion, even if the person feels fine.

Activity 2.1 — Physiology Research Tasks

~15 min

Question 2.1a — Physiological Sequence

3 marks

Put the following events in the correct order from 1 (first) to 5 (last):

EVENTS TO ORDER

cardiac arrest | submersion | laryngospasm | hypoxia | cerebral hypoxia

Order	Event	What it means
1 (first)	_____	Person goes underwater
2	_____	Airway closes involuntarily
3	_____	Oxygen levels in blood drop
4	_____	Heart stops pumping effectively
5 (last)	_____	Brain cells start dying — time-critical!

Question 2.1b — Why Start CPR Quickly?

3 marks

Use the sentence starters below to explain the 4-minute window in plain language.

When the heart stops, the brain stops getting _____ blood.

After about _____ minutes without oxygen, brain cells start _____ .

CPR generates about ___% of normal cardiac output, which is _____ to keep the brain alive.

This is why starting CPR within 4 minutes is so important:

_____

Question 2.1c — Secondary Drowning `2 marks`

Explain secondary drowning and why it matters. Use the word bank.

WORD BANK

hours later | fluid in the lungs | coughing | medical review | near-drowning | rare but serious

Secondary drowning is when _____ develops _____ a near-drowning incident.

It is clinically important because _____

After any submersion incident, even if the person seems fine, they should _____ .

Source URL:

_____

3 Social Determinants of Drowning Risk

Activity 3.1 — Social Determinants Analysis

Pairs · ~18 min

Social determinants are **non-medical factors** that influence health outcomes — things like where you grew up, your income, and cultural norms.

Question 3.1a `4 marks`

Choose TWO social determinants from the list below. For each, complete the argument framework.

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SOCIAL DETERMINANTS — CHOOSE 2

Gender norms (masculinity) | CALD background | Socioeconomic status | Rural / regional location | Alcohol use

Social Determinant 1: _____

HOW IT INCREASES RISK: _____

EVIDENCE / EXAMPLE: _____

PREVENTION STRATEGY: _____

Social Determinant 2: _____

HOW IT INCREASES RISK: _____

EVIDENCE / EXAMPLE: _____

PREVENTION STRATEGY: _____

Question 3.1b — Individual vs Structural

2 marks

What is the difference between an **individual** explanation and a **structural** explanation for a health problem? Give one example of each using the CALD drowning data.

DEFINITION BANK

personal choice | system-level | lack of swim access | didn't learn to swim

An individual explanation blames _____ choices.

Example: "They _____ ."

A structural explanation focuses on _____ factors.
Example: "In many countries of origin, _____."

Structural explanations are more useful for prevention because
_____

Question 3.1c — School Actions `3 marks`

Propose THREE specific, practical actions a Victorian school could take to reduce CALD drowning disproportionality.

Tip: be SPECIFIC — not just "raise awareness." Think: who, what, where, when, how?

Action 1: The school could _____

Action 2: The school could _____

Action 3: The school could _____

4 Evaluating Drowning Prevention Strategies

Activity 4.1 — Prevention Strategy Evaluation ~15 min

For each strategy, decide if the evidence is Strong, Moderate, or Limited, and explain a limitation.

EVIDENCE STRENGTH – CHOOSE ONE

Strong — backed by solid research | **Moderate — some research, some gaps** | **Limited — little evidence, or limited reach**

Strategy	Evidence strength	Key limitation (write in your own words)
4-sided pool fencing		
Mandatory swim lessons for primary school students		
Expanding patrolled beach patrols		
Warning signs at unpatrolled beaches		
Targeted CALD community swim programs		

Question 4.1a — Justify Your Priorities `4 marks`

If you had $5 million to spend on drowning prevention, which TWO strategies would you choose? Use the argument framework.

Strategy 1: _____

I WOULD CHOOSE THIS BECAUSE: _____

THE EVIDENCE SHOWS: _____

THE TRADE-OFF IS: _____

Strategy 2: _____

I WOULD CHOOSE THIS BECAUSE: _____

THE EVIDENCE SHOWS: _____

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THE TRADE-OFF IS: _____

5 International Comparison

Activity 5.1 — Research Comparison

~20 min · Devices needed

Visit each website and complete the table. Look for: program name, key safety message, and what makes them different.

WEBSITES TO VISIT

- 🇦🇺 royallifesaving.com.au
- 🇬🇧 rlss.org.uk/education
- 🇺🇸 redcross.org/take-a-class/water-safety

Question	🇦🇺 RLSSA (Australia)	🇬🇧 RLSS UK	🇺🇸 Red Cross (USA)
Main program name			
Key safety message / slogan			
One thing unique about their approach			

Question 5.1a — Should Victoria Adopt It?

3 marks

Choose ONE element from an international approach that is NOT currently used in Victoria. Should Victoria adopt it? Use the argument framework.

The element I chose: _____ **From:** _____ **(country/organisation)**

WHAT IT IS: _____

WHY IT COULD HELP IN VICTORIA:

I THINK VICTORIA SHOULD / SHOULD NOT (CIRCLE) ADOPT IT BECAUSE:

SOURCE URL:

6 Diving, Spinal Injuries & High-Risk Teenage Behaviour

📋 **KEY FACTS — KEEP THESE HANDY FOR YOUR ANSWERS**

Word Bank: cervical spine · quadriplegia · spinal cord · sandbank · tidal · sediment · false confidence · masculinity norms · peer pressure · AIHW · RLSSA · feet first · permanent disability · inland waterway

- Unsafe diving = **10%** of all spinal cord injuries and **20%** of quadriplegia cases in Australia (AIHW)
- **95%** of aquatic spinal injury patients are male — **75%+** are under 30 — **66%** suffer permanent disability
- Highest incidence of diving spinal injury: children aged **10–14**
- RLSSA: **"More people drown in inland waterways than any other location in Australia."**
- Core prevention message: **"Feet first, first time, every time."** Never dive into unknown water.
- +34% increase in drowning among 15–20 year olds post-COVID vs pre-COVID (RLSSA)

Sources: Royal Life Saving Australia (royallifesaving.com.au) · AIHW Diving Injury Report (aihw.gov.au) · Royal Perth Hospital spinal injury study

Activity 6.1 — Case Study Analysis

~18 min

CASE STUDY 1 · NSW 2025

Terrigal Wharf

Teenage boy dived from a wharf into shallow water. Helicopter evacuation to hospital. He didn't know how shallow it was under the structure.

CASE STUDY 2 · VICTORIA 2025

Torquay Front Beach

Teenage boy suffered suspected spinal injury diving at a surf beach. Airlifted to hospital. Sandbanks at surf beaches shift with every tide.

CASE STUDY 3 · NSW 2024

Bawley Beach Jetty

Young man seriously injured diving from a jetty. A sandbank had recently formed beneath the jumping spot. He had jumped there safely before.

Question 6.1a — Common Factors

3 marks

Identify two common factors across all three case studies. For each, use the framework below.

Factor 1 — Use this framework:

FACTOR: _____

HOW IT CAUSED THE INJURY (NOT JUST THAT IT WAS THERE): _____

WHICH CASE STUDY SHOWS THIS MOST CLEARLY: _____

Factor 2 — Use this framework:

FACTOR: _____

HOW IT CAUSED THE INJURY: _____

WHICH CASE STUDY SHOWS THIS: _____

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Question 6.1b — Why 95% of Patients Are Male

4 marks

Use the social determinants framework from Section 3. Don't just say "males take more risks." Explain WHY they take more risks using social factors.

Argument framework:

SOCIAL NORM / EXPECTATION THAT DRIVES RISK-TAKING:

HOW THIS LEADS SPECIFICALLY TO DIVING BEHAVIOUR:

WHY TELLING INDIVIDUALS TO "BE CAREFUL" IS NOT ENOUGH:

WHAT STRUCTURAL CHANGE WOULD ADDRESS THE ROOT CAUSE:

Question 6.1c — Why Neck Injuries Cause Permanent Disability

3 marks

Use the physiology content from Section 2 to explain why a cervical (neck) spine injury from diving often causes permanent quadriplegia.

Sentence starters:

"The cervical spine is responsible for _____. When a diver hits the bottom head-first, the vertebrae _____. This is different from other spine levels because _____. The result is often permanent because _____."

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Activity 6.2 — Myth vs Evidence

~15 min

Use the sentence starters below. Replace the blanks with your own words using facts from the key data section above.

Myth: "If other people are jumping there, it must be safe."

Starter: "This is a myth because depth can change due to _____ and _____. The Bawley Beach case shows that a _____ had formed between visits, making a safe spot suddenly dangerous."

Myth: "I've jumped here before — I know this spot."

Starter: "Prior experience creates false confidence because _____ changes between visits. The correct approach is to _____."

Myth: "Good swimmers are safe from diving injuries."

Starter: "Swimming ability is irrelevant to diving injuries because the injury happens _____ before swimming matters. Australian data shows _____% of patients were healthy young males."

Myth: "Jumping feet-first is safe — you won't hit your head."

Starter: "Feet-first jumping into shallow water can still cause _____ and _____. The key prevention message is _____."

Activity 6.3 — Prevention Strategy Design

~15 min

My Prevention Strategy

Step 1 — Identify your target group:

TARGET GROUP (BE SPECIFIC — AGE, GENDER, LOCATION):

STATISTIC SHOWING THIS GROUP IS AT HIGHER RISK (CITE SOURCE):

SECOND STATISTIC:

Step 2 — Describe your strategy:

WHAT WILL YOU DO:

WHERE / THROUGH WHAT CHANNEL:

WHO DELIVERS IT:

SMART METRIC — HOW WILL YOU KNOW IF IT WORKED:

Step 3 — Limitation:

THE BIGGEST WEAKNESS OF MY STRATEGY IS:

A MORE COMPLETE RESPONSE WOULD INCLUDE:

7 Assessment Task Planning

Activity 6.1 — Plan Your Assessment Task

Assessment preparation

Choose your option and use the planning scaffold to get started. Remember: all options need at least THREE sources with URLs and at least ONE statistic.

Option A — Policy Brief
600–800 words to Victorian Minister. Two recommendations + evidence + success metrics.

Option B — Campaign
Target ONE high-risk group. Campaign name, message, visual, platform, 100-word justification.

Option C — School Audit ★
Audit your school. Benchmark against Curriculum 2.0. Priority matrix + recommendations.

My Planning Notes

OPTION CHOSEN:

WHO AM I WRITING FOR (AUDIENCE / ADDRESSEE)?

MY KEY ARGUMENT / CENTRAL CLAIM:

STATISTIC I WILL USE:

SOURCE 1 URL: _____

SOURCE 2 URL: _____

SOURCE 3 URL: _____

My draft — write here or attach on a separate page:

✏️ How Did I Go? Circle: 😕 Not yet / 🙂 Getting there / 😊 Got it!

I can calculate Victorian drowning rates and compare them to other benchmarks	😕	🙂	😊
I can explain the physiological sequence of drowning and why CPR timing matters	😕	🙂	😊
I can identify social determinants of drowning risk and explain how they work	😕	🙂	😊
I can evaluate prevention strategies and justify my priorities with evidence	😕	🙂	😊
I can compare at least two international water safety approaches	😕	🙂	😊
I have a clear plan for my assessment task with at least 3 sources	😕	🙂	😊

One thing I still want help with:

_____

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